Motherboards and Architecture -- Guided Notes

Unit 2.2 | CompTIA Network+ N10-009 Obj. Core 1 3.5 | N10-008

Name: ________________________________ Date: ___________ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Match each motherboard form factor to its dimensions:

ATX: _______ mm x _______ mm

Micro-ATX: _______ mm x _______ mm

Mini-ITX: _______ mm x _______ mm

A smaller motherboard will/will not fit in a larger ATX case.

Question 2

Socket LGA 1700 is compatible with Intel _______ through _______ generation Core processors. Socket AM4 is compatible with AMD Ryzen _______ through _______ series. A CPU from one socket family _______ (will/will not) fit a motherboard with a different socket.

Question 3

DDR4 and DDR5 RAM modules are physically _______ with each other. A DDR5 stick _______ (will/will not) physically fit in a DDR4 slot because the _______ (alignment notch) is in a different position. You must match the RAM generation to the _______ specification.

Question 4

By default, RAM operates at its base _______ (e.g., 4800 MHz for DDR5-4800), not its advertised rated speed. To enable the rated speed, you must enable _______ (Intel) or _______ (AMD) in the UEFI settings. Without this, faster RAM sticks run slower than their marketed specification.

Question 5

Modern motherboards use _______ firmware instead of the legacy BIOS. UEFI supports the _______ partition table, which allows drives larger than _______ TB and up to 128 primary partitions. Legacy BIOS uses _______ which limits drives to 2 TB.

Question 6

Secure Boot is a UEFI feature that checks the _______ signature of the OS bootloader before allowing it to run. Its purpose is to prevent _______ (malware that infects the boot process) from loading. Secure Boot must be enabled for Windows _______ installation.

Question 7

The CMOS battery (typically a _______ coin cell) powers a small chip that stores the BIOS/UEFI settings when the system is unplugged. If this battery dies, the system will _______ its BIOS settings and reset the _______. Clearing the CMOS (by removing the battery or using the CLRTC jumper) resets all BIOS settings to _______.

Question 8 -- Real World Application

REAL WORLD: A customer bought a motherboard (LGA 1700, DDR5 slots) and plans to install a 13th Gen Intel Core i7, two sticks of DDR5-6000 RAM, and a 4 TB NVMe SSD. They want to run Windows 11. List every UEFI setting they should verify before and after installing Windows 11, and explain what could go wrong if each setting is wrong.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/2-2-motherboards-and-architecture/2-2-motherboards-and-architecture-answer-key.pdf
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